A note on AI: We are aware that all of these tasks can be solved by AI within seconds. However, we encourage you to work on the tasks on your own first and, whenever you get stuck, ask us (or AI) for help. If you just let AI write the solutions, you won't deeply understand what you are doing – and we will notice this directly when we ask you to present your work.

In this sheet, you organize your own small machine learning competition. Choose (at least) **one** of the three data sets (Handball, UEFA Euro football, or Tennis) and work with it throughout the whole sheet. Briefly explain your choice.

Load the Handball, UEFA Euro football, and Tennis data into your current R session.

1 Machine Learning Competition

1. Split your data into training and test data. Think about reasonable splitting strategies for your data set and justify your choice (recall Sheet 1: why should the data not simply be split randomly?). Possible approaches are, e.g., a single split by time (the last tournament, season, or years as test data), a K-fold cross-validation where each fold corresponds to one tournament or season, or a rolling window.

2. Fit at least **four** different methods to the training data, e.g., (Poisson, linear, or logistic) regression, regression or classification trees (CARTs; `rpart`), random forests (`cforest` from the `party` package or `ranger`), Extreme Gradient Boosting (XGBoost; `xgboost` package), and neural networks (e.g., the `nnet` package). Tune the hyperparameters of the methods where appropriate (see Sheet 2).

3. Think about several measures to evaluate the predictive performance of the methods. Depending on your response, these could be, e.g.,

   - the mean absolute error (MAE) or root mean squared error (RMSE) of the predicted goals or goal differences,
   - the ratio of correctly predicted match outcomes,
   - the Brier score, the log-loss, or the Rank Probability Score (RPS, see the R function below) of the predicted outcome probabilities,
   - the return of a betting strategy based on your predicted probabilities and the betting odds (tennis data: `B365.1` and `B365.2`, see the R function below).

   For the Euro data, the predicted numbers of goals of both teams can be transformed into probabilities for a win, a draw, and a loss using the Skellam distribution (`dskellam` and `pskellam` from the `skellam` package). Discuss the advantages and disadvantages of focusing on this new type of response and consider additional, suitable measures.

4. Determine the winner of your competition. Does the winner depend on the chosen measure?

```
# ----- Rank Probability Score (RPS) -----
# pred_matrix: matrix of predicted probabilities (rows = matches,
#              columns = ordered outcomes, e.g. win / draw / loss)
# obs: factor of observed outcomes (levels in the same order as the columns)
rank_prob_score <- function(pred_matrix, obs) {
  n <- nrow(pred_matrix)
  C <- ncol(pred_matrix)

  # Convert observed outcomes into one-hot matrix
  obs_mat <- matrix(0, n, C)
```

```
  obs_mat[cbind(1:n, as.integer(obs))] <- 1

  # Cumulative sums
  P_cum <- t(apply(pred_matrix, 1, cumsum))
  O_cum <- t(apply(obs_mat, 1, cumsum))

  # RPS = mean squared difference of cumulative distributions
  mean(rowSums((P_cum - O_cum)^2) / (C - 1))
}

# ----- Return of a betting strategy (tennis data) -----
# pred_probs: predicted probabilities that Player 1 wins
# odds: matrix with columns = odds for Player 1, Player 2
# outcomes: vector of true results (1 = Player 1 wins, 2 = Player 2 wins)
# tau: only bet if the expected return exceeds tau
evaluate_betting <- function(pred_probs, odds, outcomes, tau = 0) {
  # Build probability matrix [P1_win, P2_win]
  probs <- cbind(pred_probs, 1 - pred_probs)

  # Expected return: pi * odds - 1
  exp_ret <- probs * odds - 1

  # Best outcome for each match
  max_ret <- apply(exp_ret, 1, max)
  best_outcome <- apply(exp_ret, 1, which.max)  # 1 = P1, 2 = P2

  # Only bet if expected return > tau
  bet <- max_ret > tau

  # Realized profit/loss
  realized <- ifelse(
    bet & best_outcome == outcomes, odds[cbind(1:nrow(odds), outcomes)] - 1,
    ifelse(bet, -1, 0)
  )

  data.frame(
    match = 1:nrow(odds),
    best_outcome = best_outcome,
    expected_return = max_ret,
    bet = bet,
    outcome = outcomes,
    realized = realized
  )
}
```

2 Feature Engineering

1. Think about new covariates that could improve the predictions and add them to your dataset. Examples are information on the coach of a team (football), the travel distance of the away team (handball), or the head-to-head record of the two players (tennis). This may require collecting additional data, e.g., from the internet. Please document your data sources. Make sure that the new covariates only use information that was available **before** the respective match.

2. Repeat Sections 1 with the extended data set (using the same split and the same measures). Compare the results: Do the new covariates improve the predictive performance? Does the winner of your competition change?

3 Final Model and Interpretation

1. Fit (at least) your winner model to the **complete** data set (i.e., training and test data). Why does this make sense? Why can you no longer use these data to evaluate the predictive performance?

2. Interpret the final models: Which covariates are important, and how do they affect the response? Use suitable tools, e.g., the estimated coefficients (regression models), plots of the trees, or variable importance measures (random forests). You might also need to involve additional techniques from "interpretable ML" (IML) / "explainable AI".